Harvest Ledger: Crop Supply Scorecard

Dataset and dashboard guide | ITC C307-401I Foundation of Business Analytics | Mendez & Subala

1. Who the dashboard is for

A dashboard is built for **one specific audience** and shows only what that audience needs, like the dashboard of a car, which differs from the dashboard of a sports car or an F1 car. Ours is built for an **agriculture policy planner**: someone at a state or national agriculture department who decides where to focus support.

Decisions the planner makes

- Which crops is our supply too dependent on?
- Which states carry the supply, and which can be relied on year after year?
- Is production growing, and does the growth come from more land or from better yield?
- Where should land or support be shifted to raise productivity?

Everything else (detailed distributions, correlations, raw record browsing) was left out on purpose. Those belong to the midterm exploratory analysis in the notebook, not to a planner's dashboard.

2. About the dataset

India Crop Production, State Wise records how much of each crop was **planted (Area)** and **harvested (Production)** in every district of India, year by year, from **1997 to 2015**. It comes from India's Ministry of Agriculture & Farmers Welfare, shared on Kaggle (AravindPCoder). Each row answers: *in this district, year and season, how much of this crop was planted and produced?*

Item	Value
Raw rows	246,091
Columns	7 (State, District, Crop_Year, Season, Crop, Area, Production)
Years	1997 to 2015 (19 crop years)
States / districts / crops	33 / 646 / 124
Seasons	Kharif, Rabi, Whole Year, Summer, Winter, Autumn
Rows after cleaning	242,361

Units

Production is mostly in **tonnes** (1 tonne = 1,000 kg). **Coconut is counted in nuts.** The dataset itself does not state units, so totals across crops describe **recorded volume**, not weight or value. This is the most important caution in the whole project.

3. Data cleaning

Check	Result	Decision
Missing values	Only Production: 3,730 rows	Removed, leaving 242,361 rows
Duplicate rows	0	Nothing to remove
Zero Production	3,523 rows	Kept: valid recorded values
Zero or negative Area	0	Nothing to fix

Check	Result	Decision
Text spacing	Names such as 'Coconut ' had spaces	Trimmed

Raw has **more** rows than Cleaned because cleaning only removes rows. The dashboard lets you switch between Cleaned and Raw.

4. The six KPIs on the Scorecard

Each KPI answers one planner question and carries a status label. The cut-offs are **our own rules of thumb** and can be adjusted after discussing them with the professor.

KPI	What it measures	Rating rule	Value (all data)
Total production	Recorded volume for the selected years	No rating	141.18B
Annual growth	Compound growth of yearly totals, 1998 to 2014 (sparse edge years skipped)	Within +/-2% a year is Flat	+2.5% (Growing)
Top-crop share	Share of production from the single largest crop	Over 50% High dependence; 30 to 50% Moderate	92.1% Coconut (High dependence)
Top-3 state share	Share from the three largest states	Over 75% Highly concentrated; 50 to 75% Concentrated	90.2% Kerala, Andhra Pradesh, Tamil Nadu
Crops for 80% of output	How many crops, largest first, reach 80% of production	3 or fewer Very narrow; 4 to 10 Narrow	1 of 124 crops (Very narrow)
Output stability	Variation (CV) of yearly totals; lower is steadier	Under 0.15 Steady; 0.15 to 0.30 Variable; over 0.30 Volatile	0.26 (Variable)

With **Exclude coconut** switched on, the dependence and concentration KPIs change, which shows how much one crop drives the picture.

5. Dashboard map

Page	Question it answers	Visuals
Scorecard	How much, how fast, how exposed?	6 KPI cards, key takeaways, production through time, top crops, top states
Trends	Is output rising, from land or from yield?	Total production by year, yield and area
Crops	Which crops carry the supply?	Top crops, concentration (Pareto), top crops by yield
Regions	Which states carry it, and are they reliable?	Top states, output stability
Seasons	When is it harvested?	Production by season, which crops fill each season
Predict yield	How well can yield be predicted?	Model vs baseline, actual vs predicted, what drives yield, error by crop
Recommend	Where should land be shifted?	Gain by state, crop plan per state
Data notes	Can the numbers be trusted?	Raw vs cleaned, completeness, fields, method, limitations

Filters (Cleaned or Raw, years, State, Season, Crop, Show Top 5/10/15/20, Exclude coconut) apply to the first six pages. Every chart has a Chart or Table switch and a written **how to read it**, **what it means**

now and **keep in mind** note.

6. Advanced model: predictive and prescriptive

The notebook (Sections 15 and 16) adds one model on top of the descriptive work, and the dashboard shows its results on the Predict yield and Recommend pages. NLP was not used because the dataset has no free text.

Predictive: predicting crop yield

Target: yield = Production / Area (log transformed, because yield is extremely skewed). **Inputs:** state, crop, season, crop year and log area. **Model:** Gradient Boosting (HistGradientBoostingRegressor). **Test:** trained on 1997 to 2010 (190,535 records) and tested on 2011 to 2014 (51,265 records), years it never saw. A baseline that guesses each crop's average yield is used for comparison.

Measure	Baseline (crop average)	Gradient Boosting
R2	0.728	0.788
MAE (log)	0.323	0.198
RMSE (log)	0.552	0.487

- The model cuts the average error by **39%** against the baseline.
- **Typical error** for main crops is about 17% (sugarcane) to 30% (cotton), so predictions are for ranking and comparing, not exact forecasts.
- **Crop** is by far the most important feature, then **State**. Season and area add very little, and **crop year adds nothing**, so there is no steady yield trend in 1997 to 2014.
- Most of the accuracy comes from knowing which crop it is, so the baseline is already strong. The model's extra value comes from the state.

Prescriptive: where to shift land

Using the model's predicted 2015 yield for each state, season and crop, a **yield index** is built: predicted yield divided by the typical (median) yield of the same crop across states. Above 1 means the state is expected to out-yield the typical state. An index makes crops with different units comparable.

A **linear program** per state and season then maximizes the area-weighted yield index, subject to: total area fixed, at most 10% of the land moved, each crop changing by at most 25%, and only crops with an index of 1.10 or more may gain area while only crops at 0.90 or less may lose area. Coconut is excluded because it is perennial.

- **Result:** 767 state-season-crop rows across 25 states: 102 Expand, 112 Reduce, 553 Hold.
- **Largest estimated gain:** Kerala (about 11.6%), then Meghalaya (about 6.0%) and Arunachal Pradesh (about 4.8%).
- **Example, Gujarat (about 3.4% gain):** expand castor seed, banana, onion, potato, sesamum, garlic and arhar/tur; reduce Kharif rice and maize, and Whole Year sugarcane.
- Before expanding a crop, check its **yield variation**: above about 0.3 means yields have swung a lot.

Limits of the model

- There are **no prices, costs, soil, water or climate data**, so 'better' means higher predicted yield than other states, **not higher profit**.
- The plan assumes crops of the same season can replace each other, which may not hold everywhere.

- The 10% and 25% limits and the 1.10 / 0.90 thresholds are our assumptions. The output is decision support for a planner to review, not an instruction.

7. Key findings

- Recorded production is **extremely concentrated**: coconut is about 92% (roughly 23 times sugarcane) and Kerala about 69%.
- Production grew about **2.5% a year** between 1998 and 2014 and peaked in 2011. Yearly output is **variable** (CV 0.26).
- The data is strongly right-skewed: the mean production per record (about 582 thousand) is far above the median (729).
- Area and production are strongly related by rank (Spearman about 0.89) but not linearly (Pearson about 0.04), because a few huge records distort the linear measure.
- Yield depends mainly on the crop and the state, and shows no clear trend over time.

8. Limitations

- Mixed units (coconut in nuts) make cross-crop totals not strictly comparable.
- The first and last crop years are sparsely reported (1997: 8,899 records; 2015: 561; 2002 and 2003: about 17,000), so growth and trends at the edges can mislead. Growth and stability skip them.
- Rows with missing Production were dropped, which could bias totals if they are not missing at random.
- The model and recommendation results are computed on the full cleaned dataset and do not change with the dashboard filters.

9. Project files

File	Purpose
index.html, styles.css, app.js	The dashboard (HTML, CSS, JavaScript with Plotly.js)
data.js	The cleaned and raw data, embedded so no file selection is needed
model.js	Model and recommendation results exported from the notebook
plotly.min.js	Charting library, stored locally so the dashboard works offline
Group9_1_..._FINAL.ipynb	Notebook: descriptive analytics plus Sections 15 and 16 (predictive and prescriptive)
crop_production.csv	Original raw dataset